

AI Ecosystem

A clear, simple introduction to Artificial Intelligence, Machine Learning, Deep Learning, and Data Science — and how they fit together. By the end of this lesson, you will know which field a problem belongs to before you choose a tool.

L1

WHAT'S INSIDE

- 01** Four fields, defined plainly
- 02** Seven decades of AI history
- 03** How the fields nest inside each other
- 04** Recipes vs. learning by example
- 05** Where it leads — careers & tools

WHY THIS LESSON MATTERS

Four words you'll hear every day — only three of them are interchangeable

AI

~70 yr

Years since the term was coined at Dartmouth, 1956.

ML

Subset

Machine Learning lives inside AI — one method among many.

DL

Subset of
ML

Deep Learning is one method
inside Machine Learning.

DATA SCIENCE

Sister

Overlaps with all three, but isn't a subset of any.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Define AI, ML, DL and Data Science in your own words.

2

Explain how the four fields are related, with the nested diagram.

3

Pick the right field for any everyday problem in under ten seconds.

4

Spot three common mix-ups that confuse most beginners.

5

Map a real Indian system to the field that drives its result.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The four fields

Plain-language definitions of AI, ML, DL and Data Science — one example each.

02

How they fit together

DL inside ML, ML inside AI, Data Science overlapping all three.

03

Recipes vs. learning

The single analogy that separates traditional code from Machine Learning.

04

Where it leads

Real Indian systems and the career paths that grow from each field.

01

PART 1 OF 4

The four fields, defined plainly

One slide per field — definition on the left, real-world example on the right.

AI is the umbrella — about 70 years of it

Artificial Intelligence is the broadest of the four fields. It is any technique that makes a machine perform a task that would normally need human-like thinking — perceiving, deciding, responding.

AI includes everything below: rule-based systems, expert systems, Machine Learning, Deep Learning. The umbrella covers them all.

DEFINITION

Artificial Intelligence

Making machines do tasks that need human-like thinking — from simple rules to advanced learning.

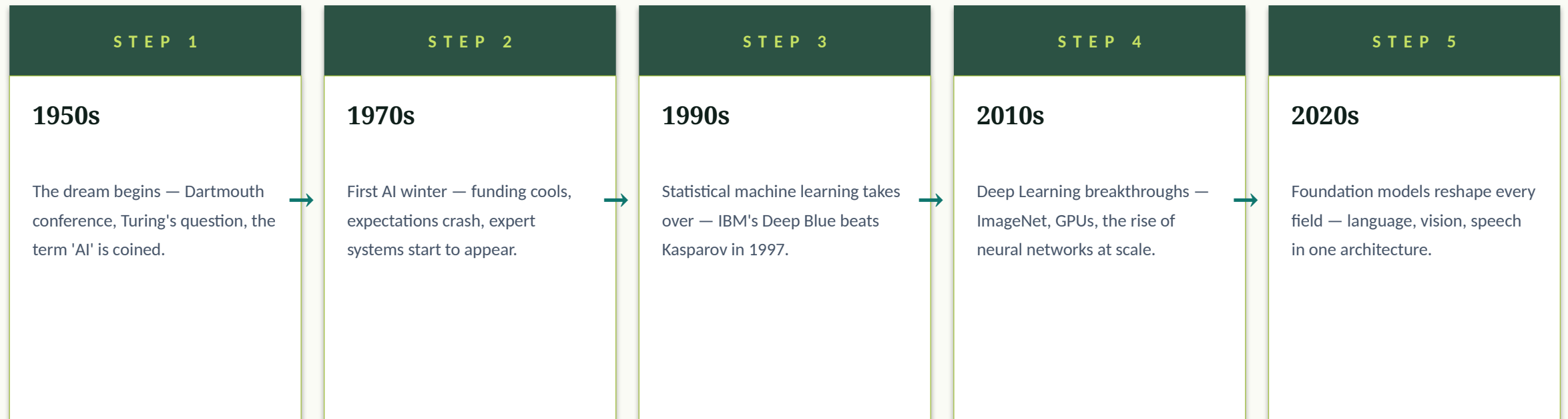
EXAMPLE

A traffic-light system that switches signals based on programmed rules, with no data or learning, is still AI.

A SHORT HISTORY OF AI

Seven decades, three winters, one revolution

Each era built the foundation the next era stood on. The names changed; the ambition did not.



ML lives inside AI — and is the workhorse you'll meet most

Machine Learning is one method inside AI. Instead of being told the rules, the system finds the rules from examples.

Most AI in industry today is Machine Learning — fraud detection, recommendation, demand forecasting, churn prediction. If a project feeds a model labelled examples, it is ML.

DEFINITION

Machine Learning

Computers that learn patterns from many examples, instead of following written rules.

EXAMPLE

A credit-card fraud model trained on 500 million past transactions to flag suspicious new ones in real time.

DL lives inside ML — the technology behind today's breakthroughs

Deep Learning is one method **inside** Machine Learning. It uses many-layered neural networks to learn from very large amounts of raw data — pixels, audio, text — without hand-crafted features.

DL is what makes face unlock, voice assistants, and modern translation possible. It is part of ML, not a replacement for it.

DEFINITION

Deep Learning

A type of Machine Learning that uses deep neural networks. Works best on large, complex data like images and text.

EXAMPLE

Aadhaar face-match unlocking your phone, or Google Translate converting Marathi audio to English text.

Data Science is a sister field — overlapping, but separate

Data Science is about turning raw data into business decisions. It uses statistics, dashboards, SQL, sometimes ML — but its purpose is decision support, not building autonomous systems.

A monthly sales dashboard with no model training is Data Science. A churn-prediction model that scores every customer is ML. The two fields overlap, but the outputs are different.

DEFINITION

Data Science

Turning data into decisions using statistics, dashboards, and queries.

EXAMPLE

A Power BI dashboard tracking last quarter's sales, built with SQL and statistics — no model training, no neural net.

02

PART 2 OF 4

How the four fields fit together

One nested diagram, one worked example, one cheat-sheet table — and the boundaries get crystal clear.

THE NESTING

Deep Learning $\subset$ Machine Learning $\subset$ AI • Data Science overlaps all three

AI is the outer ring — every method in this lesson lives somewhere inside it.

Inside AI sits Machine Learning. Inside ML sits Deep Learning.

Data Science is a separate circle that overlaps with all three — it shares tools and people, but it is its own profession with its own deliverables.

FIVE RULES OF THUMB

- If it learns from data → it's at least ML.
- If it learns from raw data via deep nets → it's DL.
- If it produces decisions, not predictions → it's Data Science.
- If it's rule-based with no data → it's AI but not ML.
- Two of the four can describe one project — name the dominant one.

WORKED EXAMPLE

One real Indian system — and where each of the four fields shows up

THE SCENARIO

Aadhaar face-match. A citizen taps their thumb at a kirana store; their face is captured by a CCTV camera; the system verifies identity and approves a payment. All four fields are present in this single workflow — once you know what to look for.

THE TAKEAWAY

AI is the umbrella. ML is the model that learns identity from examples. DL is the deep neural net that processes the face image. Data Science is the dashboard that monitors success rates by district every morning.

WHAT TO NOTICE

- 1 AI — the umbrella under which all of this lives.
- 2 ML — the model trained on millions of face pairs.
- 3 DL — the deep net that turns pixels into a face embedding.
- 4 Data Science — the dashboards tracking success by district.

QUICK REFERENCE

AI vs. ML vs. DL — at a glance

Six attributes, three fields. Use this as a cheat sheet whenever you need to choose between approaches.

Attribute	AI	ML	DL
Scope	Broadest	Subset of AI	Subset of ML
Data hunger	Optional	Moderate	Very high
Compute	Low to moderate	Moderate	GPU / TPU
Tasks	Rules, decisions	Predict, classify	Vision, speech, language
Output	A decision	A prediction	Embeddings + predictions

03

PART 3 OF 4

Recipes vs. learning by example

The single analogy that separates traditional code from Machine Learning.

THE ANALOGY

Traditional code is a recipe. Machine Learning is learning to cook.

Traditional code is like a recipe: you write the steps, the computer follows them, every time, exactly the same.

Machine Learning is the opposite: you give the computer many examples — 'these are good cakes, these are bad' — and let it figure out the steps for itself. The recipe is *learned*, not written. That is the entire shift in one sentence.

TRADITIONAL CODE

Like writing a recipe

Programmer writes the rules. Computer follows them. Same input → same output, every single time.

MACHINE LEARNING

Like learning by example

Programmer shows examples. Computer learns the rules from data. Patterns emerge that no human wrote.

Five systems you use — and the field driving each

Several use more than one field. The question is always: which one drives the user-visible result?



UPI

UPI fraud blocking

ML — gradient-boosted trees decide in <100 ms whether a transaction is fraud. AI is the umbrella.



AADHAAR

Face-match unlock

DL — a deep neural net turns face pixels into an embedding and matches against record.



FLIPKART

BBD dynamic pricing

ML — LightGBM retrained hourly across crores of SKUs. Data Science dashboards track lift.



OLA

Ride matching

AI + ML — optimisation finds the cheapest pairing; ML predicts ETAs and surge demand.



NSE

Risk dashboards

Data Science — SQL, statistics and Power BI. No model training; pure decision support.



SWIGGY

Personalised home

DL inside ML — embeddings rank dishes you'll order tonight. Sits inside the AI umbrella.

04

PART 4 OF 4

Where this leads — careers and tools

Each field has its own typical tools, job roles and applications. This is what you'd see on a resume.

Four fields, four career tracks — and the tools each one lives on

Most professionals start broad, find what they like, and specialise over two to three years.

AI

AI ENGINEER

Tools you'll see

Python, rule engines, optimisation libraries, LLM toolchains. Job titles: AI Engineer, Solutions Architect.

ML

ML ENGINEER

Tools you'll see

scikit-learn, XGBoost, LightGBM, MLflow, AWS SageMaker. Job titles: ML Engineer, Applied Scientist.

DL

DL ENGINEER

Tools you'll see

PyTorch, TensorFlow, GPUs, HuggingFace. Job titles: Research Scientist, Computer Vision Engineer.

DS

DATA SCIENTIST

Tools you'll see

SQL, pandas, Power BI / Tableau, statistics. Job titles: Data Scientist, Analytics Manager.

✦ THINK • PAIR •
SHARE

ACTIVITY

Sort five everyday systems into the right field

PROMPT

For each of these, decide which field actually does the work the user sees — AI, ML, DL or Data Science. One word each. Justify in one line.

1. *Spotify recommending you a new song*
2. *A traffic light that switches every 60 seconds, no sensors*
3. *Your phone unlocking with face recognition*
4. *A monthly sales report built in Excel and SQL*
5. *A Hindi-to-English translation app on your phone*

HOW TO RUN IT

- 1 60s solo — sort each system in your notebook.
- 2 60s with a partner — defend any borderline calls.
- 3 60s share-out — instructor calls on a team.
- 4 30s reveal — instructor reveals the canonical sort.

A bank wants to flag suspicious credit-card transactions in real time, with every decision auditable. Which approach fits best?

A

Rule-based AI with a hand-tuned threshold list

B

Classical ML with gradient-boosted trees

C

A large language model with chain-of-thought prompting

D

Pure Data Science — a Power BI dashboard

A bank wants to flag suspicious credit-card transactions in real time, with every decision auditable. Which approach fits best?

CORRECT ANSWER



Classical ML with gradient-boosted trees

WHY

Latency under 100 ms and per-decision audibility are the two design constraints. GBM serves both — fast inference and feature importance per call. Rule-based AI can't keep up with novel fraud; LLMs fail latency and audit; a dashboard doesn't make per-transaction calls.

A monthly sales dashboard built with SQL queries and statistics — no model training, no neural net — is best classified as:

A

Artificial Intelligence

B

Machine Learning

C

Deep Learning

D

Data Science

A monthly sales dashboard built with SQL queries and statistics — no model training, no neural net — is best classified as:

CORRECT ANSWER

**Data Science****WHY**

No model is being trained, so it's not ML or DL. It isn't a decision-making machine in the AI sense either. It is decision support built on data — the textbook definition of a Data Science deliverable.

END OF LESSON 1

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Define AI, ML, DL and Data Science in your own words.
- 2 Draw the nested diagram and place Data Science correctly.
- 3 Pick the right field for any everyday system in under 10 seconds.
- 4 Spot three common mix-ups and correct them in conversation.
- 5 Map a real Indian system to the field driving its user-visible result.

NEXT → LESSON 2 — Industry Applications of AI